Abstract Algebra by Pinter, Chapter 31

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Chapter 31 on Galois Theory Preamble

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Note that root field is called splitting field in other texts.

A. Examples of Root Fields over $\mathbb{Q}$

Q1

```
sage: solve(x^2 - 2*x - 2, x)
[x == -sqrt(3) + 1, x == sqrt(3) + 1]
sage: solve(x^2 + 1, x)
[x == -I, x == I]
```

Q2

Possible roots for $x^2 - 3$ are $\pm 1, \pm 3$.

```
sage: p = lambda x: x^2 - 3
sage: p(-1), p(1), p(-3), p(3)
(-2, -2, 6, 6)
```

So it is irreducible.

For $x^2 - 2x - 2$ we can use Eisenstein's irreducibility criteria.

```
sage: solve(x^2 - 3, x)
[x == -sqrt(3), x == sqrt(3)]
sage: solve(x^2 - 2*x - 2, x)
[x == -sqrt(3) + 1, x == sqrt(3) + 1]
```

Root field is $\mathbb{Q}(\sqrt{3})$.

Q3

```
sage: solve(x^4 - 2, x)
[x == I*2^(1/4), x == -2^(1/4), x == -I*2^(1/4), x == 2^(1/4)]
```

Therefore the root field is $\mathbb{Q}(2^{\frac{1}{4}}, i)$

Since $\mathbb{Q}(2^{\frac{1}{4}}) \subseteq \mathbb{R}$, then the root field of $x^4 - 2$ over $\mathbb{R}$ is $\mathbb{R}(i)$.

Q4

```
sage: solve(x^4 - 2*x^2 + 9, x)
[x == -sqrt(2*I*sqrt(2) + 1), x == sqrt(2*I*sqrt(2) + 1), x == -sqrt(-2*I*sqrt(2) + 1), x == sqrt(-2*I*
```

Root field is $\mathbb{Q}(i, \sqrt{2})$.

```
sage: solve(x^2 - 2*sqrt(2)*x + 3, x)
[x == sqrt(2) - I, x == sqrt(2) + I]
```

Root field is $\mathbb{Q}(i, \sqrt{2})$.

Q5

```
sage: c
sqrt(3) + I
sage: ((c^2 - 2)^2).expand()
-12
sage: ((x^2 - 2)^2
....: ).expand()
x^4 - 4*x^2 + 4
sage: x^4 - 4*x^2 + 4 + 12
x^4 - 4*x^2 + 16
sage: solve(x^4 - 4*x^2 + 4 + 12, x)
[x == -sqrt(2*I*sqrt(3) + 2), x == sqrt(2*I*sqrt(3) + 2), x == -sqrt(-2*I*sqrt(3) + 2), x == sqrt(-2*I*
```

So there are roots $-2\sqrt{3}i - 2$ and $2\sqrt{3}i - 2$.

And $x^2 - 3$ for $b(x)$.

```
sage: c = sqrt(2) + sqrt(3)
sage: (c^2).expand()
2*sqrt(3)*sqrt(2) + 5
sage: (((c^2).expand() - 5)^2).expand()
24
sage: ((x^2 - 5)^2).expand()
x^4 - 10*x^2 + 25
sage: solve(x^4 - 10*x^2 + 1, x)
[x == -sqrt(2*sqrt(6) + 5), x == sqrt(2*sqrt(6) + 5), x == -sqrt(-2*sqrt(6) + 5), x == sqrt(-2*sqrt(6)
```

Remembering that $\sqrt{6} = \sqrt{2}\sqrt{3}$.

Q6

All of them are valid root fields except the cube root one.

B. Examples of Root Fields over $\mathbb{Z}_p$

Q1

```
sage: R.<x> = IntegerModRing(3)[]
sage: f = x^3 + 2*x + 1
sage: S.<u> = R.extension(f)
sage: (x - u)*(x - (u + 1))*(x - (u + 2))
x^3 + 2*x + 1
```

List the elements:

```
sage: S = R.quotient(x^3 + 2*x + 1, 'u')
sage: len(S)
27
sage: list(S)
[0,
 1,
 2,
 u,
 u + 1,
 u + 2,
 2*u,
 2*u + 1,
 2*u + 2,
 u^2,
 u^2 + 1,
 u^2 + 2,
 u^2 + u,
 u^2 + u + 1,
 u^2 + u + 2,
 u^2 + 2*u,
 u^2 + 2*u + 1,
 u^2 + 2*u + 2,
 2*u^2,
 2*u^2 + 1,
 2*u^2 + 2,
 2*u^2 + u,
 2*u^2 + u + 1,
 2*u^2 + u + 2,
 2*u^2 + 2*u,
 2*u^2 + 2*u + 1,
 2*u^2 + 2*u + 2]
```

Roots of $a(x)$ are $u, u+1, u+2$.

Root field is therefore $\mathbb{Z}_3(u)$.

Q2

```
sage: S = R.quotient(x^2 + x + 2, 'u')
sage: list(S)
[0, 1, 2, u, u + 1, u + 2, 2*u, 2*u + 1, 2*u + 2]
```

Utilize the fact that $u^2 = -u - 2 = 2u + 1$, we can make the addition and multiplication tables.

```
sage: from sage.matrix.operation_table import OperationTable
sage: OperationTable(S, operation=operator.add)
+  a b c d e f g h i
 +------------------
a| a b c d e f g h i
b| b c a e f d h i g
c| c a b f d e i g h
d| d e f g h i a b c
```

```
e| e f d h i g b c a
f| f d e i g h c a b
g| g h i a b c d e f
h| h i g b c a e f d
i| i g h c a b f d e
```

```
sage: OperationTable(S, operation=operator.mul)
*  a b c d e f g h i
 +------------------
a| a a a a a a a a a
b| a b c d e f g h i
c| a c b g i h d f e
d| a d g h b e f i c
e| a e i b f g c d h
f| a f h e g c i b d
g| a g d f c i h e b
h| a h f i d b e c g
i| a i e c h d b g f
```

Roots are: $u, 2u + 2$ (figured by substitution)

Root field: $\mathbb{Z}_3(u)$

Q3

Root field will be all combos of polynomials over $\mathbb{Z}_2$ of degree 2 polynomials.

```
sage: R.<x> = IntegerModRing(2)[]
sage: S = R.quotient(x^3 + x^2 + 1, 'u')
sage: list(S)
[0, 1, u, u + 1, u^2, u^2 + 1, u^2 + u, u^2 + u + 1]
```

In $\mathbb{Z}_2$, $-1 = 1$ so $u^3 = u^2 + 1$.

```
sage: OperationTable(S, operation=operator.add)
+  a b c d e f g h
 +----------------
a| a b c d e f g h
b| b a d c f e h g
c| c d a b g h e f
d| d c b a h g f e
e| e f g h a b c d
f| f e h g b a d c
g| g h e f c d a b
h| h g f e d c b a
```

```
sage: OperationTable(S, operation=operator.mul)
*  a b c d e f g h
 +----------------
a| a a a a a a a a
b| a b c d e f g h
c| a c e g f h b d
d| a d g f b c h e
e| a e f b h d c g
f| a f h c d g e b
g| a g b h c e d f
h| a h d e g b f c
```

```
sage: R.<x> = IntegerModRing(2)[]
sage: f = x^3 + x^2 + 1
sage: S.<u> = R.extension(f)
...
sage: f(u)
0
```

```
sage: f(u^2)
0
sage: f(u^2 + u + 1)
0
sage: (x - u)*(x - u^2)*(x - (u^2 + u + 1))
x^3 + x^2 + 1
```

Root field: $\mathbb{Z}_3(u)$

Q4

```
sage: f = x^3 + x + 1
sage: S.<u> = R.extension(f)
sage: f(u)
0
sage: f(u + 1)
u^2 + u
sage: f(u^2)
0
sage: f(u^2 + 1)
u
sage: f(u^2 + u)
0
sage: f(u^2 + u + 1)
u^2
```

Roots are $u, u^2, u^2 + u$.

Q5

```
sage: R.<x> = IntegerModRing(3)[]
sage: f = x^3 + x^2 + x + 2
sage: S.<u> = R.extension(f)
sage: f(u)
0
sage: f(u + 1)
2*u
sage: f(u + 2)
u + 2
sage: f(u^2)
u^2
sage: f(u^2 + 1)
0
sage: f(u^2 + 2)
2*u^2 + 2
sage: f(u^2 + u)
2*u^2 + u
sage: f(u^2 + u + 1)
u^2
sage: f(u^2 + 2*u)
2*u^2 + 2*u
sage: f(u^2 + 2*u + 1)
u^2
sage: f(u^2 + u + 2)
2*u + 2
sage: f(u^2 + 2*u + 2)
u + 2
sage: f(2*u)
2*u^2 + 1
sage: f(2*u + 1)
2*u^2 + u + 1
sage: f(2*u + 2)
```

```
2*u^2 + 2*u
sage: f(2*u^2)
2*u^2 + u + 2
sage: f(2*u^2 + 1)
u + 2
sage: f(2*u^2 + 2)
u^2 + u + 1
sage: f(2*u^2 + u)
u^2 + 1
sage: f(2*u^2 + 2*u)
2*u^2 + 2*u
sage: f(2*u^2 + u + 1)
2*u^2 + 2*u + 1
sage: f(2*u^2 + u + 2)
u
sage: f(2*u^2 + 2*u + 1)
0
```

I think the answer in the book is wrong since

```
sage: R.<x> = IntegerModRing(3)[]
sage: f = x^3 + x^2 + x + 2
sage: S.<u> = R.extension(f)
sage: u^3 + u^2 + u + 2
0
sage: f(u^2 + 1)
0
sage: f(2*u^2 + 2*u + 1)
0
sage: (x - u)*(x - (u^2 + 1))*(x - (2*u^2 + 2*u + 1))
x^3 + x^2 + x + 2
```

We can even substitute these roots into $q(x)$ which is claimed to be irreducible

```
sage: a = lambda x: (x^2 + (u + 1)*x + (u^2 + u + 1))
sage: a(u^2 + 1)
0
sage: a(2*u^2 + 2*u + 1)
0
```

Roots are: $u, u^2+1, 2u^2+2u+1$.

Basis is $\{u^2, u, 1\}$.

C. Short Questions Relating to Root Field

Q1

The basis for a degree 2 extension $\mathbb{F}(c)$ is $\{c, 1\}$ with $c^2 \in \mathbb{F}$. This includes $-c$ and so $\mathbb{F}(c)$ is the root field for $(x+c)(x-c) = x^2 - c^2$.

The question I assume is asking to prove every degree 2 field extension is normal. So since $c \in F(c)$, then we can divide the polynomial by $(x-c)$, leaving a degree 1 polynomial $(x-\alpha) \in F(c)[x]$ or that $\alpha \in F(c)$.

Q2

$a(x)$ as a polynomial of degree n over a finite field will have n distinct roots. All roots $c_1, \ldots, c_n \in K$ and form a root field over F. Assume some roots are in I, then the roots of $a(x)$ are still the same, and so K forms a root field over I as well.

Q3

Any polynomial can be factored into a linear combination of complex roots. This means $\mathbb{C}$ is the root field of every polynomial. Since polynomials can also have root fields contained in $\mathbb{R}$, and $\mathbb{R} \subset \mathbb{C}$, this means the root

field is either $\mathbb{R}$ or $\mathbb{C}$ of every polynomial.

Q4

This question is impossible. See the answer here.

Q5

As per the back of the book, we just need to show that $F(d_1, d_2) = F(c)$ and that it's the splitting field.

$$\begin{aligned} d_1^2 &= \frac{a^2}{4} - b \\ d_2^2 &= \frac{a}{2} + d_1 \\ &= \frac{a}{2} + \sqrt{\frac{a^2}{4} - b} \end{aligned}$$

Roots of $p(x)$ are $\pm\sqrt{\frac{a}{2} \pm d_1}$. We know that $\frac{a}{2} \pm d_1 \in F(d_1)$.

$[F(d_1) : F] = 2$. Every extension of degree 2 is a root field from 31C1 above.

Q6

$$\sigma_c(a(x)) = a(c)$$
$$\ker \sigma_c = \{a(x) : a(c) = 0\}$$

Every ideal of a field is principal, so $J = \langle p(x) \rangle$. For some $a(x) \in J$, $a(x)$ is a multiple of $p(x)$.

$$\sigma_c : F[x] \to F$$

so $F(c)$ contains $p(x)$ since the ideal J contains all polynomials with c as their root, and all polys are a multiple of $p(x)$.

Q7

From the previous argument $p(x)$ is the lowest degree polynomial in J and hence irreducible. Otherwise $p(c) = f(c)g(c) = 0 \implies f(c) = 0$ or $g(c) = 0$ which would be a contradiction since it implies there would be a lower degree polynomial in J. Every root field is a simple extension $F(c)$, we can convert a polynomial with roots a, b to one with c from theorem 2 in this chapter.

Q8

Give an isomorphism $h(x)$ which fixes F, and a polynomial

$$a(x) = a_0 + a_1 x + \cdots + a_n x^n$$

We can see that where c is an algebraic root in K, then

$$\begin{aligned} a(h(c)) &= a_0 + a_1 h(c) + \cdots + a_n h(c^n) \\ &= h(a(c)) = h(0) = 0 \end{aligned}$$

So also $h(c)$ is a root of $a(x)$ and the isomorphism simply permutes roots since it sends unique elements of K to K'. We also observe that the mapping is one to one, fixing F, permuting roots, and that $h : K \to K$.

$$h(\{c_1, \ldots, c_n\}) = \{c_1, \ldots, c_n\}$$

Since the polynomial $a(x)$ is irreducible over F, and K being a finite extension can be reduced to a simple extension $F(c) = K$, where c is a root of $a(x)$, which means that

$$F(c) \cong F/\langle a(x) \rangle$$

which forms a vector space of $\deg a(x) = n \implies [K : F] = n$.

Given another polynomial $b(x)$ where $b(c) = 0 : c \in K \implies b(h(c)) = 0 \implies$ K is the splitting field for $b(x)$. This means $b(x)$ is split completely by K into linear factors.

Note, this means every polynomial of degree n which has a root in K makes K its splitting field. The converse does not hold. An irreducible polynomial of degree n does not necessarily have a splitting field of degree n. See this answer.

Q9

First we prove this for an irreducible polynomial $p(x)$ with $n = \deg p(x)$ roots of the form $c_1, \dots, c_n$. Inductively adjoining c_1 to F forms a field $F(c_1)$ such that $[F(c_1) : F] = n$ with a basis $\{1, c_1, \dots, c_1^{n-1}\}$. Dividing $p(x)$ by $(x - c_1)$ leaves a polynomial $q(x)$ with degree $n - 1$ and adjoining the second root to $F(c_1)$ forms a field extension with degree $[F(c_1, c_2) : F(c_1)] = n - 1$. Proceeding in this way, we obtain a maximum order of $n!$.

The polynomial $a(x)$ is reducible to n irreducible factors $a(x) = p_1(x)p_2(x)\cdots p_n(x)$ where $\deg p_i(x) = k_i$. Each of these fields forms a simple extension $F(c_1, \cdots, c_n)$, where $[F(c_1, \cdots, c_n) : F] = [F(c_1, \cdots, c_n) : F(c_1, \cdots, c_{n-1})] \cdots [F(c_1) : F] = k_n \cdot k_{n-1} \cdots k_1$.

Since $\deg a(x) = N$ and $k_1 + \cdots k_n$, so $[K : F] \mid k_1! \cdots k_n!$.

From the binomial formula, given $n = k + l$, then there's an integer z such that

$$z = \frac{n!}{k!l!} \implies n! = zk!l!$$

So therefore given $n = k_1 + \cdots + k_n$, we can see that $x \mid k_1! \cdots k_n! \implies x \mid n!$

D. Reducing Iterated Extensions to Simple Extensions

Q1

a

$$\mathbb{Q}(\sqrt{2}, i\sqrt{3}) = \mathbb{Q}(\sqrt{2} + i\sqrt{3})$$
$$c = \sqrt{2} + i\sqrt{3}$$
$$[\mathbb{Q}(c) : \mathbb{Q}] = 4$$
$$\implies \sqrt{2} = a_0 + a_1 c + a_2 c^2 + a_3 c^3$$
$$c^2 = 2i\sqrt{6} - 1 \implies i\sqrt{6} \in \mathbb{Q}(c)$$
$$c^3 = 4i\sqrt{3} - \sqrt{2} - 6\sqrt{2} - i\sqrt{3} = 3i\sqrt{3} - 7\sqrt{2}$$
$$c^3 + 7c = 10i\sqrt{3} \implies i\sqrt{3} \in \mathbb{Q}(c)$$

Same can be shown for $\sqrt{2}$.

b

$$(\sqrt[6]{2})^3 = \sqrt{2}, (\sqrt[6]{2})^2 = \sqrt[3]{2}$$
$$\implies \mathbb{Q}(\sqrt{2}, \sqrt[3]{2}) = \mathbb{Q}(\sqrt[6]{2})$$

Q2

The roots of $x^2 - 2x - 1$ by completing the square are $\pm\sqrt{2} + 1$.

For a cubic with real coefficients in a field, it either has all real roots or 2 complex roots. By differentiating and sketching the curve where it's increasing or decreasing, we see that this cubic has two complex roots.

According to the **Complex Conjugate Theorem**, if $x = a + ib$ is a solution to a polynomial with real coefficients, then so is $x = a - ib$.

Thus we conclude that $\mathbb{Q}(a, b) = \mathbb{Q}(a + b)$ where a is a complex root of $x^3 - x - 1$.

Q3

These factors are all linearly independent so $c = \sqrt{2} + \sqrt{3} + \sqrt{-5}$.

Q4

From C6, because $\sqrt{2}$ and $\sqrt{3}$ are independent, the basis is $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$ and so the minimum polynomial has degree 4.

```
sage: # We let x = sqrt(2) + sqrt(3), so now we square it on both sides
sage: (sqrt(2) + sqrt(3))^2
(sqrt(3) + sqrt(2))^2
sage: ((sqrt(2) + sqrt(3))^2).expand()
2*sqrt(3)*sqrt(2) + 5
sage: # so now (x^2 - 5) = 2*sqrt(3)*sqrt(2)
sage: # lets square again both sides
sage: (2*sqrt(3)*sqrt(2))**2
24
sage: ((x^2 - 5)^2).expand()
x^4 - 10*x^2 + 25
sage: ((x^2 - 5)^2).expand() - 24
x^4 - 10*x^2 + 1
```

Q5

The minimum polynomial will now have degree 8.

```
sage: x - sqrt(2) == sqrt(3) + I*sqrt(5)
x - sqrt(2) == I*sqrt(5) + sqrt(3)
sage: (x - sqrt(2) == sqrt(3) + I*sqrt(5))^2
(x - sqrt(2))^2 == (I*sqrt(5) + sqrt(3))^2
sage: ((x - sqrt(2) == sqrt(3) + I*sqrt(5))^2).expand()
x^2 - 2*sqrt(2)*x + 2 == 2*I*sqrt(5)*sqrt(3) - 2
sage: ((x - sqrt(2) == sqrt(3) + I*sqrt(5))^2).expand() + 2*sqrt(2)*x
x^2 + 2 == 2*sqrt(2)*x + 2*I*sqrt(5)*sqrt(3) - 2
sage: p = ((x - sqrt(2) == sqrt(3) + I*sqrt(5))^2).expand() + 2*sqrt(2)*x
sage: p
x^2 + 2 == 2*sqrt(2)*x + 2*I*sqrt(5)*sqrt(3) - 2
sage: p += 2
sage: p
x^2 + 4 == 2*sqrt(2)*x + 2*I*sqrt(5)*sqrt(3)
sage: (p^2).expand()
x^4 + 8*x^2 + 16 == 8*I*sqrt(5)*sqrt(3)*sqrt(2)*x + 8*x^2 - 60
sage: (p^2).expand() + 60 - 8*x^2
x^4 + 76 == 8*I*sqrt(5)*sqrt(3)*sqrt(2)*x
sage: p = (p^2).expand() + 60 - 8*x^2
sage: (p^2).expand()
x^8 + 152*x^4 + 5776 == -1920*x^2
sage: (p^2).expand() + 1920*x^2
x^8 + 152*x^4 + 1920*x^2 + 5776 == 0
sage: p = x^8 + 152*x^4 + 1920*x^2 + 5776
sage: p(x = sqrt(2) + sqrt(3) + I*sqrt(5)).expand()
0
```

E. Roots of Unity and Radical Extensions

Q1

The roots of $x^n - 1$ are $1, \omega, \ldots, \omega^{n-1}$ which is the basis for $\mathbb{Q}(\omega)$ generated by ω since it is primitive.

Q2

Define a substitution function σ_ω

$$\sigma_\omega(a(x)) = a(\omega)$$

σ_ω is a homomorphism because

$$\begin{aligned}\sigma_\omega(a(x)b(x)) &= a(\omega)b(\omega)\\ &= \sigma_\omega(a(x))\sigma_\omega(b(x))\end{aligned}$$

Which has a kernel of

$$\begin{aligned}\ker\sigma_\omega &= \{a(x) : \sigma_\omega(a(x)) = a(\omega) = 0\}\\ &= J\end{aligned}$$

The kernel of any homomorphism is an ideal. In $F[x]$ every ideal is a principal ideal so $J = \langle p(x)\rangle$. So $p(x)$ is a polynomial of lowest degree among all nonzero polynomials in J. Hence it is irreducible.

When n is prime, then

$$x^{n-1} + x^{n-2} + \cdots + x + 1$$

is irreducible. Therefore $p(x) = x^{n-1} + \cdots + 1$ and $p(\omega) = 0$. Since

$$\mathbb{Q}(\omega) \cong \mathbb{Q}[x]/\langle p(x)\rangle$$

Then

$$[\mathbb{Q}(\omega) : \mathbb{Q}] = \deg p(x) = n - 1$$

Q3

$$p(\omega) = 0 \implies \omega^{n-1} = -(\omega^{n-2} + \cdots + \omega + 1)$$

Q4

n = 6

$$x^6 - 1 = (x^3 - 1)(x^3 + 1)$$

The roots are $1, s, s^2$ and $-1, -s, -s^2$ respectively where s is the third root of unity.

But from above we know that $s^2 + s + 1 = 0 \implies s^2 = -(s+1)$ and $s^2 \in \mathbb{Q}(s)$, so $\mathbb{Q}(\omega) = \mathbb{Q}(s)$ with basis $\{1, s\}$.

$$\begin{aligned}[\mathbb{Q}(\omega) : \mathbb{Q}] &= [\mathbb{Q}(s) : \mathbb{Q}]\\ &= 2\end{aligned}$$

n = 7

$n = 7$ is prime so $[\mathbb{Q}(\omega) : \mathbb{Q}] = 6$.

n = 8

$$\begin{aligned}x^8 - 1 &= (x^4 - 1)(x^4 + 1)\\ &= (x^2 - 1)(x^2 + 1)(x^4 + 1)\end{aligned}$$

With roots $-1, 1$ and $i, -i$ for $(x^2 - 1)$ and $(x^2 + 1)$ respectively.

The 4th roots of -1 are $\pm\frac{1}{\sqrt{2}} \pm \frac{i}{\sqrt{2}}$.

$\mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{2}, i)$ with a basis $\{1, \sqrt{2}, i, \sqrt{2}i\}$ and

$$[\mathbb{Q}(\omega) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, i) : \mathbb{Q}] = 4$$

Q5

$$\forall r \in \{1, 2, \ldots, n-1\}, (\sqrt[n]{a}\omega^r)^n = 1$$

$$|\{\sqrt[n]{a}\omega^r : r \in \{0, 1, \ldots, n-1\}\}| = n$$

Q6

The basis of $\mathbb{Q}(\omega, \sqrt[n]{a})$ is the set $\{\omega^i(\sqrt[n]{a})^j\}$ where $i, j \in \{0, 1, \dots, n-1\}$, which contains the ideal for $\sigma(c) = x^n - 1$, which is $J = \{\sqrt[n]{a}, \sqrt[n]{a}\omega, \dots, \sqrt[n]{a}\omega^{n-1}\}$.

Q7

The degree of $[\mathbb{Q}(\omega, \sqrt[3]{2}) : \mathbb{Q}] = [\mathbb{Q}(\omega, \sqrt[3]{2}) : \mathbb{Q}(\sqrt[3]{2})][\mathbb{Q}(\sqrt[3]{2} : \mathbb{Q}] = 2 \times 3 = 6$.

You can calculate $\cos(\pi/3)$ and $\sin(\pi/3)$ by splitting an equilateral triangle with unit sides in half. The sum of a triangle's angles will always be π, and so each corner of the equilateral triangle will be $(\pi/3)$. Use pythagoreas theorem to calculate the midline as $o^2 = 1^2 - (\frac{1}{2})^2$.

Using wolfram alpha, we see that the cube roots of 1 are $1, -\frac{1}{2} + \frac{\sqrt{3}}{2}i, -\frac{1}{2} - \frac{\sqrt{3}}{2}i$.

$$\mathbb{Q}(\omega, \sqrt[3]{2}) = \mathbb{Q}(i\sqrt{3}, \sqrt[3]{2})$$

Q8

Let s be the nth root of a in K, then it is the root of $x^n - a$. Then $\sqrt[n]{a}\omega^i$ is also a root of this polynomial. For every n there is an irreducible cyclotomic polynomial $p(x)$ with roots that consist of $\phi(n)$ nth roots of a that have a multiplicative order of n. By theorem 7, since $p(x)$ has one root in K and is irreducible, therefore all its roots are in K.

For n is not prime, $p(x) \mid x^n - a$ so all $\phi(n)$ primitive roots of $p(x)$ are also roots of $x^n - a$ and so generates all roots. We use the irreducible polynomial $p(x)$ to prove there's an isomorphism permuting roots of $p(x)$, therefore showing both fields are equivalent and contain all roots.

Fixing $\mathbb{Q}$, there is an isomorphism $h : \mathbb{Q}(s) \to \mathbb{Q}(\sqrt[n]{a}\omega^i)$. Note that $p(h(c)) = (h(c))^n - a = h(c^n - a) = 0$ as the function is homomorphic so $h(c)^n = h(c^n)$ and fixes $\mathbb{Q}$ leaving a unchanged. This function is an isomorphism mapping one to one and onto, we can say that $h(x)$ permutes the roots c of $x^n - a$. So we can see that $\mathbb{Q}(s) = \mathbb{Q}(\sqrt[n]{a}\omega^i)$, and that K contains all roots of $x^n - a$.

Since the spltting field contains all powers of $\sqrt[n]{a}\omega^i$ by the isomorphism $h(x)$ as well as $\sqrt[n]{a}\omega^0 = \sqrt[n]{a}$ and also its inverse $(\sqrt[n]{a})^{-1}$ (because it's a field), so it also contains all nth roots of unity since $(\sqrt[n]{a})^{-1}(\sqrt[n]{a}\omega^i) = \omega^i$.

F. Separable and Inseparable Polynomials

Q1

From theorem 1, F has characteristic $0 \implies$ irreducible polynomials never have multiple roots.

Q2

Each power of x in $a(x)$ is independent, so $a_m x^m + a_n x^n \neq 0$ when $m \neq 0$ unless $a_m = a_n = 0$.

Therefore $a'(x) = \sum m_i a_i x^{m_i - 1} = 0 \implies p \mid m_i$ for all m_i and all nonzero terms of $a(x)$ are of the form $a_{mp}x^{mp}$.

Q3

$$a(x) = (x - c)^2 q(x)$$

$$a'(x) = 2(x - c)q(x) + (x - c)^2 q'(x)$$

Both $a(x)$ and $a'(x)$ have c as a root, but $a(x)$ is irreducible, so $a(x) \mid a'(x)$, but this cannot be true since $\deg a'(x) < \deg a(x)$ unless $a'(x) = 0$.

From the previous answer $a'(x) = 0$ means the nonzero terms of $a(x)$ are of the form $a_{mp}x^{mp}$, and $a(x)$ is a polynomial in powers of x^p.

Q4

$a(x)$ is a polynomial in powers of $x^p \implies a'(x) = 0 \implies a(x) \mid a'(x) \implies$ share common factor of $a(x) \implies a(x)$ has a multiple root.

Q5

This follows from $[a(x)+b(x)]^p = a(x)^p + b(x)^p$ in any field of characteristic p and is proved in 24D6.

Q6

$$a(x) = a_0 + a_1 x + \cdots + a_n x^n$$
$$a(x^p) = a_0 + a_1 x^p + \cdots + a_n x^{np}$$

The frobenius automorphism for a finite field of characteristic p is bijective since the function is injective, and any injective function from a finite set to itself is also surjective.

This implies the coefficients of $a(x^p)$ all have pth roots, and so

$$\begin{aligned} a(x^p) &= c_0^p + c_1^p x^p + \cdots + c_n^p x^{np} \\ &= (c_0 + c_1 x + \cdots + c_n x^n)^p \end{aligned}$$

Thus $b(x) = c_0 + c_1 x + \cdots + c_n x^n$ and $a(x^p) = [b(x)]^p$.

Q7

Assume there is an irreducible polynomial $a(x)$ that is inseperable. Then it is a polynomial in powers of x^p.

Then there is a polynomial $[b(x)]^p = a(x)$. Thus $a(x)$ is reducible, which is a contradiction.

Thus every irreducible polynomial is separable.

G. Multiple Roots over Infinite Fields of Nonzero Characteristic

Q1

There are infinite powers of y, so $\mathbb{Z}_p[y]$ is an infinite ring with characteristic p.

Thus

$$\mathbb{Z}_p(y) = \{a(y)/b(y) : a(y), b(y) \in \mathbb{Z}_p[y]\}$$

is an infinite field, and so is $\mathbb{Z}_p(y^p)$

Q2

By binomial theorem, all coefficients for the terms of $(x-y)^p$ are a factor of p. $E[x]$ has characteristic p so

$$x^p - y^p = (x-y)^p$$

However $y \notin K[x]$ so $x^p - y^p$ is irreducible in $K[x]$.

Q3

$$\bar{i}(a_0 + \cdots + a_n x^n) = i(a_0) + \cdots + i(a_n)x^n$$

but for all $a_i \in F, i(a_i) = a_i$ so $\bar{i}(p(x)) = p(x)$.

Q4

Expanding out $(x-a)^m$, the coefficients in F and x remain fixed, but a is mapped to b, so

$$\bar{i}((x-a)^m) = (x-b)^m$$

Q5

Since $\bar{i}$ leaves $p(x)$ fixed, so

$$\begin{aligned} \bar{i}(p(x)) &= \bar{i}((x-a)^m s(x)) \\ &= (x-b)^m \bar{i}(s(x)) \\ &= p(x) \end{aligned}$$

so a and b have the same multiplicity in $p(x)$.

H. An Isomorphism Extension Theorem (Proof of Theorem 3)

Q1

$$F_1(a) \cong F_1[x]/\langle p(x)\rangle$$

Where $p(x)$ is the minimum polynomial with a as a root. The homomorphism $\phi : F_1[x] \to F_1(a)$ by $\phi_c(a(x)) = a(c)$ has the kernel $J = \langle p(x)\rangle$ since in $F[x]$ every ideal is a principal ideal.

Thus since $s(x) = c(x) - d(x)$ has a root a since $s(a) = 0$, so $s(x) \in J$ and it is a multiple of $p(x)$.

Observing that $h(c(x) - d(x)) = h(p(x)q(x))$, we easily see that $h(p(x)q(x)) = hp(x)hq(x)$ and also that $h(c(x) - d(x)) = hc(x) - hd(x)$ since

$$\begin{aligned} hc(x) - hd(x) &= (h(c_0) - h(d_0)) + (h(c_1) - h(d_1))x + \cdots + (h(c_n) - h(d_n))x^n \\ &= h(c_0 - d_0) + h(c_1 - d_1)x + \cdots + h(c_n - d_n)x^n \\ &= h(c(x) - d(x)) \end{aligned}$$

Q2

$$\begin{aligned} h(c(a)) &= h(c_0) + h(c_1)b + \cdots + h(c_n)b^n \\ h(d(a)) &= h(d_0) + h(d_1)b + \cdots + h(d_n)b^n \end{aligned}$$

$$\begin{aligned} h(c(a)) - h(d(a)) &= h(c_0 - d_0) + h(c_1 - d_1)b + \cdots + h(c_n - d_n)b^n \\ &= h(c(x) - d(x))(b) \\ &= [hp(x)(b)][hq(x)(b)] \end{aligned}$$

But $hp(x)(b) = 0$ so $h(c(a)) - h(d(a)) = 0 \implies h(c(a)) = h(d(a))$.

Q3

$$hc(x) = hd(x)$$

$$\begin{aligned} \implies hc(x) - hd(x) &= 0 \\ &= h(c(x) - d(x)) \\ &= h(c_0 - d_0) + h(c_1 - d_1)x + \cdots + h(c_n - d_n)x^n \end{aligned}$$

But h is isomorphic on $F_1 \to F_2$ so $c_i = d_i \implies c(x) = d(x)$.

Q4

$h(a) = b$, there is no other value that produces b.

All the coefficients for a polynomial $c(x)$ are reversible.

$$c(x) = h^{-1}(h(c_0)) + h^{-1}(h(c_1))x + \cdots + h^{-1}(h(c_n))x^n$$

Q5

$$\begin{aligned} h(c(x) + d(x)) &= h(c_0 + d_0) + h(c_1 + d_1)x + \cdots + h(c_n + d_n)x^n \\ &= (h(c_0) + h(d_0)) + (h(c_1) + h(d_1))x + \cdots + (h(c_n) + h(d_n))x^n \\ &= hc(x) + hd(x) \end{aligned}$$

$$\begin{aligned} h(c(x)d(x)) &= h(c_0 d_0) + h(\sum_{i+j=1} c_i d_j x) + h(\sum_{i+j=2} c_i d_j x^2) + \cdots + h(\sum_{i+j=n} c_i d_j x^n) \\ &= h(c_0)h(d_0) + \sum_{i+j=1} h(c_i)h(d_j)x + \sum_{i+j=2} h(c_i)h(d_j)x^2 + \cdots + \sum_{i+j=n} h(c_i)h(d_j)x^n \\ &= hc(x)hd(x) \end{aligned}$$

I. Uniqueness of the Root Field

Q1

First note that

$$F_1(u) \cong F_1[x]/\langle p(x)\rangle$$

Let $f : F_1[x] \to F_2(v)$ defined by

$$f(a(x)) = h(a(x))(v)$$

Then the ideal is $J = \langle p(x)\rangle$

$$\begin{aligned} f(a(x)) = f(b(x)) &\iff f(a(x)) - f(b(x)) = 0 \\ &\iff f(a(x) - b(x)) = 0 \\ &\iff a(x) - b(x) \in J \\ &\iff J + a(x) = J + b(x) \end{aligned}$$

Let $\phi : F_1[x]/\langle p(x)\rangle \to F_2(v)$ by

$$\phi(J + a(x)) = f(a(x)) = h(a(x))(v)$$

We can see that this function is an ismomorphism:

- ***injective***: $\phi(J + a(x)) = \phi(J + b(x)) \implies f(a(x)) = f(b(x))$
$$\implies J + a(x) = J + b(x)$$
- ***surjective***: $h(a(x))(v) = h(a(u))$ which is **onto** $F_2(v)$ and surjective, so $f(a(x)) = \phi(J+a(x))$ is surjective.
- Finally,

$$\begin{aligned} \phi(J + a(x)) + \phi(J + b(x)) &= f(a(x)) + f(b(x)) \\ &= [ha(x) + hb(x)](v) \\ &= h(a(x) + b(x))(v) \\ &= \phi(J + a(x) + b(x)) \end{aligned}$$

$$\begin{aligned} \implies F_2(v) &\cong F_1[x]/\langle p(x)\rangle \\ F_1(u) &\cong F_1[x]/\langle p(x)\rangle \\ F_1(u) &\cong F_2(v) \end{aligned}$$

Q2

We start with $F_1(u) = K_1$ and want to prove that this means $F_2(v) = K_2$. This will also automatically prove the converse statement if shown to be true.

Let $p(x)$ be the minimum polynomial $p(x)$ for u such that $p(u) = 0$.

$h : F_1(u) \to F_2(v)$ but $F_1(u) = K_1$ so $h : K_1 \to F_2(v)$, but h is surjective and so $\deg p(x) = \deg hp(x)$ since both $p(x)$ and $hp(x)$ are irreducible.

$$\begin{aligned} \forall u_i \in K_1 : p(u_i) = 0, \exists v_i = h(u_i) : hp(v_i) = 0 \\ \implies F_2(v) = K_2 \end{aligned}$$

Because there are $\deg hp(x) = \deg p(x)$ such roots v_i which correspond to u_i roots of $p(x)$.

Q3

$$\begin{aligned} a(x) &= p(x)q(x) \\ p(u) &= 0 \end{aligned}$$

$$\begin{aligned} h(p(u)) &= h(p(x))(v) = 0 \\ &= [hp(x)](v) = hp(v) \end{aligned}$$

Since both are equivalent.

We see that v is a root of $hp(x)$.

$F_1(u) = K_1$

If $F_1(u) = K_1$, then $F_1(u)$ contains all roots of $p(x)$ and then

$$F_1(u) = K_1 \iff F_2(v) = K_2$$

Recalling that $p(x)$ is an irreducible factor of $a(x)$,

$$u \in K_1$$
$$p(u) = 0$$
$$v \in K_2$$
$$hp(v) = 0$$
$$\implies F_1(u) \cong F_2(v)$$

Putting both together

$$K_1 \cong K_2$$

$F_1(u) \neq K_1$

See that we can extend h fixing the base field.

$$h(u) = v$$
$$h : F_1(u) \to F_2(v)$$

In $F_1(u)[x]$, $a(x) = (x-u)a_1(x)$.

In $F_2(v)[x], ha(x) = (x-v)ha_1(x)$.

And $\deg a_1(x) = \deg ha_1(x) = n-1$

Now let there be a new $u' \in K_1, u' \notin F_1(u'), p(u') = 0$ and likewise for $hp(x)$ and v.

$\deg a(x) = 1$

Lastly when $n = \deg a(x) = 1$, then $K_1 = F_1$ and $K_2 = F_2$, since the basis are simply scalars and $a(x)$ is of the form $(x-a)$. The root of $a(x)$ is in F_1 itself, and h is an ismomorphism from $F_1 \to F_2$ so

$$K_1 \cong K_2$$

Q4

$$u \in K_1, v \in K_2$$

But $F_1 = F_2$

$$h : F[x] \to F[x]$$
$$\forall a \in F, h(a) = a$$
$$h = \mathrm{id}_F$$
$$h(u) = v$$
$$\implies K_1 \cong K_2$$

J. Extending Isomorphism

Q1

$$h : \mathbb{Q}(\omega) \to \mathbb{C}, \forall a \in \mathbb{Q}, h(a) = a$$

Let $b \in \mathbb{Q}(\omega)$

$$b = s_0 + s_1\omega + s_2\omega^2 + \cdots + s_{p-1}\omega^{p-1}$$
$$h(b) = s_0 + s_1h(\omega) + s_2h(\omega)^2 + \cdots + s_{p-1}\omega^{p-1}$$

So h is determined by $h(\omega)$.

Since isomorphisms preserve roots, we deduce they permute roots. There are $p-1$ roots for the minimum polynomial of ω which has degree $p-1$.

Q2

$p(x)$ is irreducible in $F[x]$. $c \in \mathbb{C} : p(c) = 0$.

Let $h : F \to \mathbb{C}$ be a monomorphism (injective homomorphism), then $h : F \to h(F)$ is an isomorphism and

$$F \cong h(F)$$

The minimum polynomial are $p(x)$ and $hp(x)$ respectively with $\deg p(x) = \deg hp(x) = n$. Since h permutes roots and by theorem 7 contains all roots, there are n possible monomorphisms.

Q3

$$h : F \to h(F)$$
$$\phi : K \to \mathbb{C}$$
$$[K : F] = n$$

So K forms a splitting field over F for a minimum polynomial $p(x) \in F[x]$ of degree n.

From the previous question we see that there are n monomorphisms $F(c) = K \to \mathbb{C}$.

Q4

$$h : \mathbb{Q} \to \mathbb{C}$$
$$h(x) = x$$
$$h(x) = h(y) \implies x = y$$
$$h(1) = 1_{\mathbb{C}}$$

$$h\left(\frac{n}{n}\right) = h\left(\frac{1}{n} + \cdots + \frac{1}{n}\right)$$
$$1 = nh\left(\frac{1}{n}\right)$$
$$\implies h\left(\frac{1}{n}\right) = \frac{1}{n}$$

$$\begin{aligned} h\left(\frac{p}{q}\right) &= h\left(\frac{1}{q} + \cdots + \frac{1}{q}\right) \\ &= \underbrace{h\left(\frac{1}{q}\right) + \cdots + h\left(\frac{1}{q}\right)}_{p} \\ &= \frac{p}{q} \end{aligned}$$

Thus all monomorphisms $h : \mathbb{Q}(a) \to \mathbb{C}$ fix $\mathbb{Q}$.

Q5

$$\mathbb{Q}(\sqrt[3]{2}) \to \mathbb{C}$$

Minimum polynomial for $c = \sqrt[3]{2}$ is $p(x) = x^3 - 2$

Three roots of $p(x)$ are $\sqrt[3]{2}$, $\sqrt[3]{2}\omega$, $\sqrt[3]{2}\omega^2$.

$\mathbb{Q}$ remains fixed. Roots are permuted (see above questions).

Three monomorphisms are

$$\sqrt[3]{2} \to \sqrt[3]{2}$$
$$\sqrt[3]{2} \to \sqrt[3]{2}\omega$$
$$\sqrt[3]{2} \to \sqrt[3]{2}\omega^2$$

K. Normal Extensions

Q1

K is a finite extension of $F \implies K$ is a simple extension and so that $K = F(c)$.

There is a minimum polynomial $p(x)$ for c in F.

So by the question K is a normal extension.

Q2

K is a finite extension of $F \implies K = F(c)$.

Let $p(x)$ be the minimum polynomial for c, so $p(c) = 0$.

Let h be an isomorphism fixing F

$$h : K \to h(K)$$

Then by the question $h(K) \subseteq K$ and since h is an isomorphism where $K \cong h(K)$, so $h(K) = K$, and $h : K \to K$ is an automorphism fixing F and permuting roots of $p(x)$.